

Final Project

Part 1: Identify three questions

Question 1: Which country in the world produces the best Arabica coffee? This requires review data on different coffees farmed all over the world.

Hypothesis: Brazil produces most coffee in the world and so do many African countries. Indonesia does also, but my impression is that Brazil and African countries would be valued higher, so I suspect that these should do well in taste tests.

Question 2: Which Arabica coffee variety is the best? This requires review data on different coffee varieties.

Hypothesis: This is a more difficult one since I do not have specific knowledge on varieties. Quick googling reveals that Bourbon and Caturra are popular Arabica species varieties. Yellow Bourbon is sold as relatively popular Brazil Bourbon Estate coffee in Finland. I suspect these varieties do well. Otherwise, this question is also about learning about varieties.

Question 3: When looking for coffees with good aftertaste, which countries or varieties should one look for? This requires more detailed taste profiling on different coffee varieties around the world.

Hypothesis: This is basically a same kind of question than 1 and 2 but now added emphasis on a more particular aspect of the coffee taste. Based on Q1 and Q2 reasoning, same hypothesis applies here as well, so Brazil and Africa in terms of geography and Caturra and Bourbon in terms of varieties.

As for the business case, these questions and research data help a hypothetical Coffee Shop or Coffee reseller/importer to make decisions on different countries and varieties based on quality and taste attributes that they want to specialize in. I chose Arabica since it is a more popular coffee species compared to Robusta.

Suitable review data to use for these questions was found from Kaggle:

<https://www.kaggle.com/datasets/ankurchavda/coffee-beans-reviews-by-coffee-quality-institute>.

This is a collection of detailed taste profiles done by Coffee Quality Institute. It is a non-profit organization to improve the quality of coffee and lives of the producers

(<https://www.coffeeinstitute.org/about>).

Part 2: Master the data

Standard Data Request Form

GENERAL INFORMATION (ALL FIELDS ARE REQUIRED)

Requester Name: Matti Pihlaja	Requester Position Title: Student
Requester Business Unit:	Requester Department: Accounting
Requester Contact Number:	Requester Email: matti.pihlaja@aalto.fi
Date Requested: 1/6/20	
Supervisor Name:	Supervisor Email:

REQUEST JUSTIFICATION (REQUIRED)

Coffee review data is needed for completion of a student project which aims at investigating the quality of coffee across varieties and countries.

REPORT DETAIL (ALL FIELDS ARE REQUIRED)

Request Description: Country.of.Origin, Variety, Acidity, Aftertaste, Aroma, Body, Balance, Flavor, Uniformity, Sweetness, Total.Cup.Points		
Report Output: ☐ PDF ☐ Excel ☒ Text (CSV) ☐ Other: _____		
List of Columns Needed in the Finished Report (ex: EMPLID, Name, Job Code, Dept ID, Dept Name) Country.of.Origin, Variety, Acidity, Aftertaste, Aroma, Body, Balance, Flavor, Uniformity, Sweetness, Total.Cup.Points		
Report Type: Summary ☐ Detail ☒	Sort Preference: Country.of.Origin	Group by Fields: No grouping

ADDITIONAL COMMENTS (CALCULATED FIELDS, SPECIAL FORMULAS OR OTHER)**REQUESTOR CONFIRMATION**

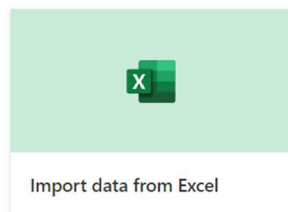
Requester Signature: _____	Date: 1/6/20
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Data dictionary

Key	Required	Attribute Name	Description	Data Type	Default Value
	N	Acidity	The quality of acidity (both high and low acidic coffees can have a high score if they perform as expected among their class)	Decimal number	null
	N	Aftertaste	Length of positive flavor	Decimal number	null
	N	Aroma	Fragrance and smell	Decimal number	null
	N	Balance	How balanced flavor, aftertaste, acidity and body are	Decimal number	null
	N	Body	Tactile feeling	Decimal number	null
	N	Country.of.Origin	Where coffee is farmed	String	null
	N	Flavor	Principal character in between first impression and aftertaste	Decimal number	null
PK	Y	ID	Row identifier	Integer	Auto-increment
	N	Sweetness	Perceived sweetness (as opposed to sourness)	Decimal number	null
	N	Total.Cup.Points	Total score	Decimal number	null
	N	Uniformity	Consistency of flavor between different cups	Decimal number	null
	N	Variety	Plant classification	String	null

Data validation and cleaning

I started with loading the input csv file Coffee-modified.csv downloaded from Kaggle in Power BI and choosing transform data to start the Power Query Editor.



On

Coffee-modified.csv

File Origin: 65001: Unicode (UTF-8) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

ID	Species	Owner	Country.of.Origin	Farm.Name	Lot.Number	Mill	ICO.
1	Arabica	metad plc	Ethiopia	metad plc		metad plc	201
2	Arabica	metad plc	Ethiopia	metad plc		metad plc	201
3	Arabica	grounds for health admin	Guatemala	san marcos barrancas "san cristobal cuch			
4	Arabica	yidnekachew dabessa	Ethiopia	yidnekachew dabessa coffee plantation		wolensu	
5	Arabica	metad plc	Ethiopia	metad plc		metad plc	201
6	Arabica	ji-ae ahn	Brazil				
7	Arabica	hugo valdivia	Peru			hvc	
8	Arabica	ethiopia commodity exchange	Ethiopia	aoime		c.p.w.e	010
9	Arabica	ethiopia commodity exchange	Ethiopia	aoime		c.p.w.e	010
10	Arabica	diamond enterprise plc	Ethiopia	tulla coffee farm		tulla coffee farm	201
11	Arabica	mohammed lalo	Ethiopia	fahem coffee plantation			
12	Arabica	cqi q coffee sample representative	United States	el filo			unk
13	Arabica	cqi q coffee sample representative	United States	los cedros			unk
14	Arabica	grounds for health admin	United States (Hawaii)	arianna farms			
15	Arabica	ethiopia commodity exchange	Ethiopia	aoime		c.p.w.e	010
16	Arabica	cqi q coffee sample representative	United States	el Ñigula			unk
17	Arabica	grounds for health admin	Indonesia	toarco jaya			010
18	Arabica	ethiopia commodity exchange					010
19	Arabica	yunnan coffee exchange	China	echo coffee	YNC-06114	echo coffee mill	100
20	Arabica	essencecoffee	Ethiopia	drima zede		drima zede	

Extract Table Using Examples | Load | Transform Data | Cancel

The original csv had many columns which were not relevant to my undertaking, so I removed those. Data types were wrong for grade columns such as Aroma which should contain decimal numbers and not text. I chose the conversion option to decimal number, but this resulted in around 10 rows. The following picture demonstrates one of these errors which show invalid data row. From Power BI I chose to remove the rows with the number error correction due to corrupt data. After this conversion succeeded and then I proceeded to check that the grading numbers for Aroma do not contain negative numbers by sorting the column in ascending order.

Species	Country.of.Origin	Company	Producer	Variety	Aroma	Flavor	Aftertaste
Specialty Coffee Association	0878a7d4b9d35d0b0fe2ce69a2...						
Arabica	Taiwan	blossom valley???	Huang san lang ???			null	
CQI Taiwan ICP CQI??????						null	
2013/2014						null	
Arabica	Brazil	klem organics coffee	NAGIPE VIANA KLEM			null	
Arabica	Guatemala					null	
Arabica	Taiwan	red on tree co., ltd.	HUANG MEI TAO ???			null	
09e3c859cac41901d54f4bd3...	m					null	
Arabica	Honduras	cigrah s.a de c.v	Reinerio Zepeda	Caturra	0 0	0	0
Arabica	Colombia	racafe & cia s.c.a		Caturra	5,08 7.75	7.83	
Arabica	Guatemala	eduardo ambroci	Conquista/Morito	Bourbon	6,17 6.5	6.5	
Arabica	Costa Rica	santa laura exportadora de cafe s.l.e.c. s.a.	Beneficio la Eva S.A.	Caturra	6,33 6.5	6.33	
Arabica	Guatemala	asociación nacional del café - anacafe -	Conquista/Morito	Bourbon	6,42 6.5	6.5	
Arabica	Mexico		RICARDO AARON SAMPIERI MARINI	Bourbon	6,5 6.67	6.42	
Arabica	Guatemala	unex guatemala, s.a.	AGRIPEC DE LA VEGA Y CIA,	Bourbon	6,5 6.33	6.5	
Arabica	Guatemala	unex guatemala, s.a.	Conquista / Morito	Bourbon	6,67 6.5	6.33	
Arabica	Mexico		Various smallholders	Typica	6,67 6.33	6.42	
Arabica	Honduras	cadessa	Omar Acosta	Catual	6,67 6.5	6.17	
Arabica	Haiti	haiti coffee	COEB Kopé			6.42	
Arabica	Mexico	café@ monteabuelo s.a. de c.v.	GONZALO			6.67	
Arabica	Guatemala	unex guatemala, s.a.	Conquista			6.17	
Arabica	Mexico		Various sm			6.83	
Arabica						6.42	
Arabica	Honduras	cafes finos de exportacion s de r. l.	Cooperativ			6.67	
Arabica	Honduras	cafes finos de exportacion s de r. l.	Cooperativ			6.67	
Arabica	Brazil	bourbon specialty coffees	Manoel Ca			6.92	
Arabica	Nicaragua	exportadora atlantic s.a				6.92	
Arabica	Philippines	prime3	various sm			6.5	
Arabica	Nicaragua	asociación aldea global jinotega	Asociación			7.17	
Arabica	Taiwan	red on tree co., ltd.	GUO JIUN HUNG ??? & TSENG RU FENG ???	Typica	6,83 6.75	7.17	
Arabica	Thailand	coffees co., ltd	Coffees Co., Ltd.	Caturra	6,83 7.42	7.42	
Arabica	Haiti	haiti coffee	Mixed	Typica	6,83 6.83	6.83	
Arabica	Myanmar		U Soe	Catual	6,92 7.5	7	
Arabica	United States (Hawaii)	kona pacific farmers cooperative		Hawaiian Kona	6,92 6.92	6.92	
Arabica	Taiwan	blossom valley????	???	Typica	6,92 7	7.17	
Arabica	Nicaragua	cooperativa el gorrión r.l	Cooperativa de Servicio Múltiples el Gorrión R.L	Caturra	6,92 6.92	7	

This revealed empty values on rows which obviously did not contain valid review data as well as one row with 0 for every reviewed category. I cleaned these by removing the rows. Then I

proceeded to repeat this process for the rest of the grade categories that should contain a decimal number: Acidity, Aftertaste, Body, Balance, Uniformity, Sweetness and Total.Cup.Points. The first round of cleaning Aroma column had removed all the offending data also for rest of the grade categories and converting these to decimal went without errors. The data also passed the check of grades having sensible non-negative values with maximum of ten. Total.Cup.Points is the exception is that contains a total number of the points.

Then I inspected Variety column and encountered both null values and a variety called “Other”. Since by my reasoning both a missing value “null” and information “Other” refer to the variety information not being available, I chose to convert these two values to string “UNSPECIFIED”. For Country I found one row with a null value for country and chose to remove that row. Here is snippet of the cleaned data in Power BI Transform, which I proceeded to load for further analysis.

	Country of Origin	Variety	1.2 Aroma	1.2 Flavor	1.2 Aftertaste	1.2 Acidity	1.2 Body	1.2 Balance	1.2 Uniformity	1.2 Sweetness	1.2 Total.Cup.Points
1	Guatemala	Catuai	7.5	6.67	6.67	7.67	7.33	6.67	8	1.33	59.83
2	Nicaragua	Caturra	7.25	6.58	6.58	6.25	6.42	6.08	6	6	63.08
3	Haiti	Typica	6.75	6.58	6.42	6.67	7.08	6.67	9.83	6	67.92
4	Mexico	Bourbon	7.08	6.83	6.25	7.42	7.25	6.75	10	10	68.33
5	Honduras	Catuai	7	6.17	6.17	6.67	6.5	6.17	8	8	69.17
6	Honduras	Catuai	7	6.33	6.17	6.5	6.67	6.17	8	8	69.17
7	Honduras	Catuai	6.67	6.5	6.17	6.67	6.83	6.17	8	8	69.33
8	Brazil	UNSPECIFIED	7	7	6.83	7	7.33	6.83	6	10	70.67
9	Mexico	Typica	6.83	7	6.83	6.92	7.42	6.92	6	10	70.75
10	Mexico	Bourbon	6.5	6.67	6.42	7.17	7.33	6.5	8	10	71
11	Mexico	Mundo Novo	6.92	6.92	6.92	6.92	7.17	7	10	10	71.08
12	Costa Rica	Caturra	6.33	6.5	6.33	6.83	6.33	6.5	10	10	71.75
13	Haiti	UNSPECIFIED	6.92	6.75	7.08	7.17	7.33	6.67	10	8.67	72.33
14	United States (Hawaii)	Hawaiian Kona	6.92	6.92	6.92	7.5	7.58	7.17	6.67	9.33	72.58
15	Colombia	UNSPECIFIED	7.33	7.33	7.33	7.42	7.67	6.67	6.67	6.67	72.83
16	Mexico	Typica	7.08	6.92	7.08	6.92	6.92	6.92	8	8	72.92
17	Honduras	Pacas	7	7.17	6.83	7	7.17	7	8	8	73.42
18	Brazil	UNSPECIFIED	7.25	6.58	6.58	7	7	6.42	8	10	73.5
19	United States (Hawaii)	Hawaiian Kona	7.17	7.17	7.17	7.68	7.67	6.92	6.67	6.67	73.67

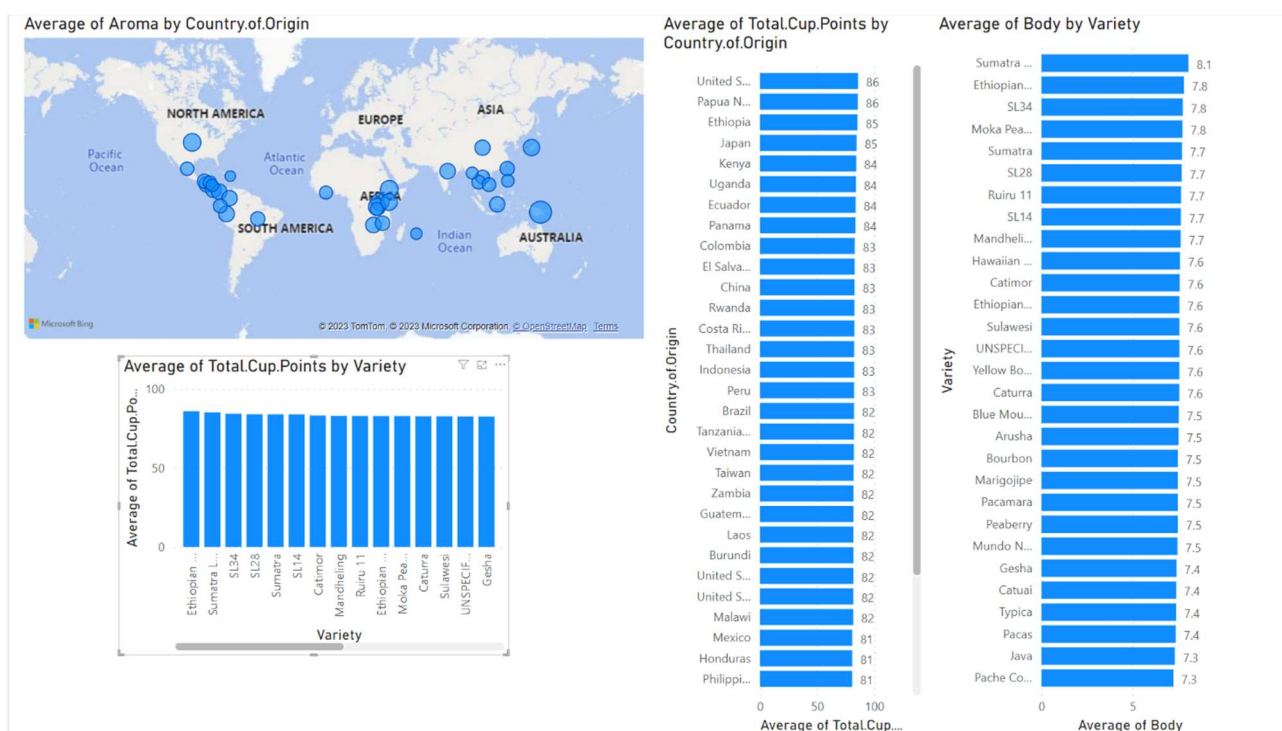
Part 3: Perform an Analysis of Data

I chose the data analytics approach of unsupervised profiling for these questions. The reason for this is that the questions are very exploratory in nature and aimed at finding different kinds of aggregate summaries based on different categories using quantitative data. Since it is exploratory the approach relies heavily on interactive visualizations done on Power BI.

The benefit of exploratory visualizations is the possibility to get a quick overview of the data and the relationship between different values and categories. This is much harder when looking at just the numbers in an excel sheet. The downside is that there is no predictive model in use to predict a question, such as predicting the review scores for a new coffee bean of Caturra variety from Colombia.

The following sections of Part 3 will describe the process of how I developed the dashboards that resulted in trying out different approaches in Power BI after I had transformed and loaded the input review data.

First, I wanted to build some charts that would show some taste category average score between varieties and countries. The process was quite straightforward, and the result is in the next picture.



The column chart creation was only a matter of choosing a score category on one axis and toggling average and choosing the dimension, either country or variety. The bubble size on the map reflects how high the average aroma is for a particular country. I found that with this data a bar chart works best and then all varieties can fit on the screen.

A problem with these charts is that one would have to create bar charts and a map for each of the score categories if one wants to examine each visually. It would not be too big of a task, but the dashboard would become rather heavy with visuals with this many score categories as there are. It would be better if the score category could be dynamic.

It turned out that Power BI Field Parameter feature can be used to accomplish this. With field parameters the user can have a slicer to toggle a field to be shown. First, I created measures to use as field parameters. I want averages so I created a measure for score category as shown below.

```
1 Avg_aftertaste = AVERAGEX('Coffee-modified', 'Coffee-modified'[Aftertaste])
```

Then I could create the slicer.

Parameters

Add parameters to visuals and DAX expressions so people can use slicers to adjust the inputs and see different outcomes. [Learn more](#)

What will your variable adjust?

Fields

Name

Parameter

Add and reorder fields

Avg_aftertaste

Avg_aroma

Avg_balance

Avg_body

☒ Add slicer to this page

Fields

Search

Coffee-modified

☐ ∑ Acidity
 ☐ ∑ Aftertaste
 ☐ ∑ Aroma
 ☐ Avg_acidity
 ☒ Avg_aftertaste
 ☒ Avg_aroma
 ☒ Avg_balance
 ☒ Avg_body
 ☐ Avg_flavor
 ☐ Avg_sweetness
 ☐ Avg_totalpoints

Create

Cancel

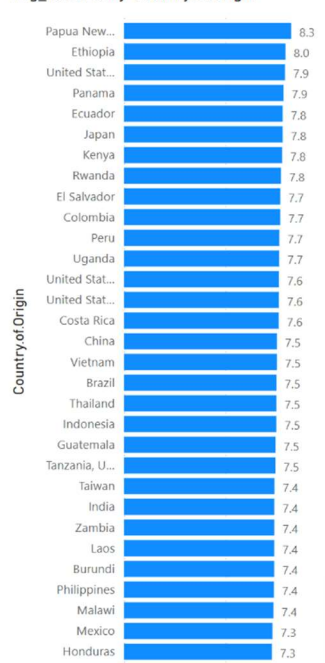
Power BI internally created a table to host the field parameters. To incorporate the dynamic field selection to the chart I changed the X-Axis for the bar charts to this new “Score Type” instead of a fixed one such as flavor. The result looks like the following.

Avg_balance by Country.of.Origin

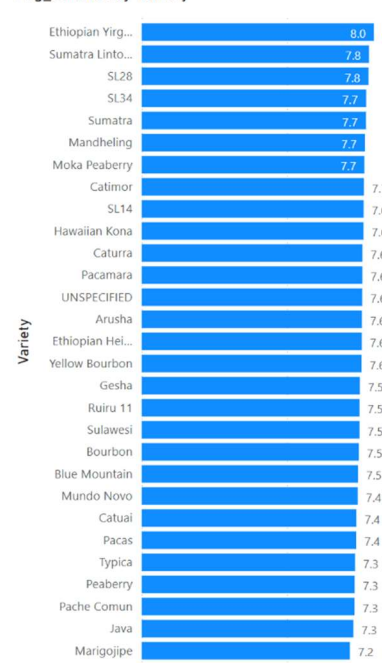


- Score Type
- ☐ Avg_acidity
 - ☐ Avg_aftertaste
 - ☐ Avg_aroma
 - ☒ Avg_balance
 - ☐ Avg_body
 - ☐ Avg_flavor
 - ☐ Avg_sweetness
 - ☐ Avg_totalpoints
 - ☐ Avg_uniformity

Avg_balance by Country.of.Origin

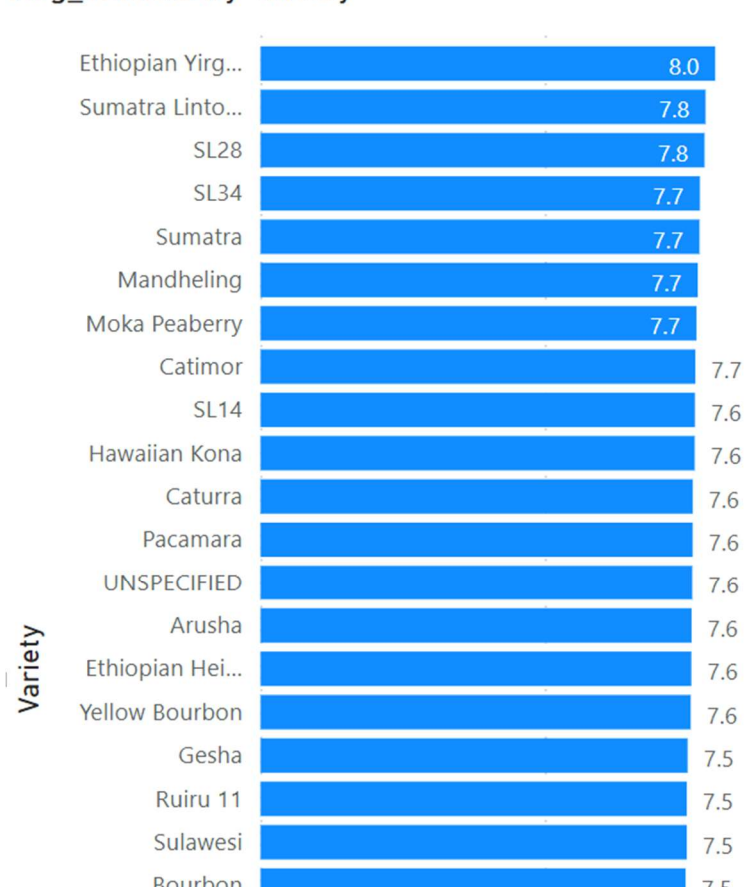


Avg_balance by Variety



All the charts now depend on the Score Type slicer and update themselves upon a user changing the selection. The data view for the right-most bar chart is depicted in the next image.

Avg_balance by Variety



Visualizations

Build visual

Filters

Y-axis: Variety

X-axis: Score Type

Legend

Add data fields here

Small multiples

Add data fields here

Tooltips

Add data fields here

Drill through

Cross-report: Off

Keep all filters: On

Add drill-through fields here

Data

Search

Coffee-modified

- ☐ Σ Acidity
- ☐ Σ Aftertaste
- ☐ Σ Aroma
- ☐ Avg_acidity
- ☐ Avg_aftertaste
- ☐ Avg_aroma
- ☐ Avg_balance
- ☐ Avg_body
- ☐ Avg_flavor
- ☐ Avg_sweetness
- ☐ Avg_totalpoints
- ☐ Avg_uniformity
- ☐ Σ Balance
- ☐ Σ Body
- ☐ Count_rows
- ☐ Country.of.Origin
- ☐ Σ Flavor
- ☐ Σ Sweetness
- ☐ Σ Total.Cup.Points
- ☐ TotalPointsRoun...
- ☐ Σ Uniformity
- ☒ Variety

Score Type

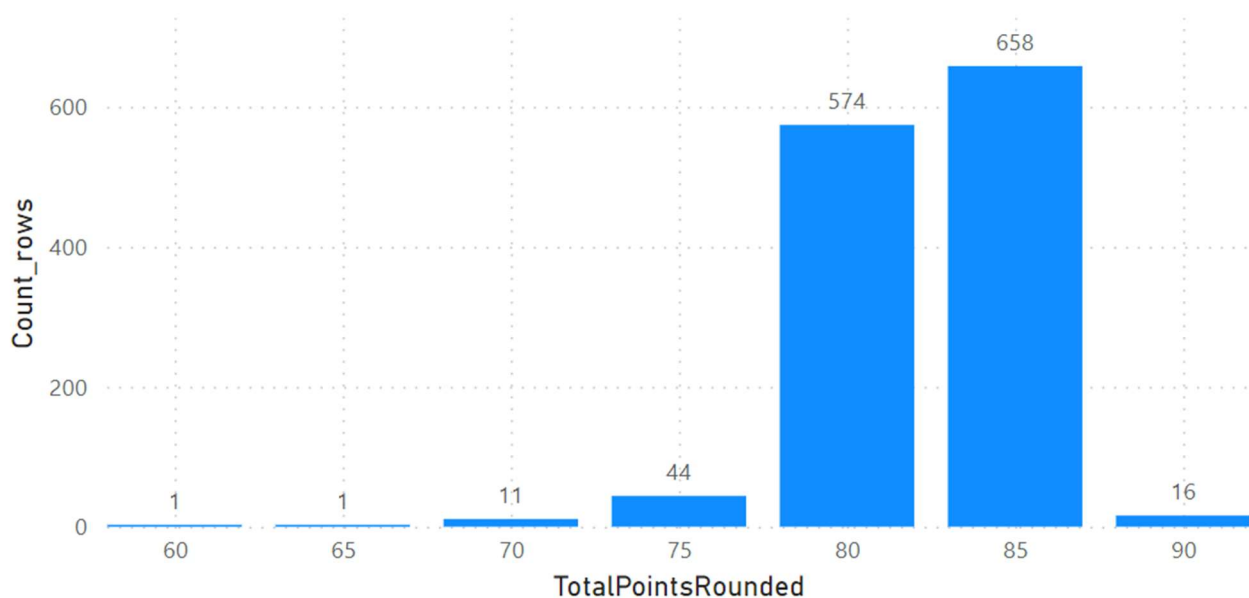
- ☒ Score Type

This shows the new “avg” measures that the Score Type field parameter is built from and the use of the Score Type field in the bar chart configuration.

I also wanted to build a couple of graphs to inspect how total points are distributed. Surprisingly there was no native histogram in Power BI, so I resorted to DIY solution. First I created a new column that rounded the Total.Cup.Points to nearest five. Then I could use this column to visualize the counts.

```
TotalPointsRounded = MROUND('Coffee-modified'[Total.Cup.Points],5)
```

Count_rows by TotalPointsRounded



The bin labels effectively mean 77,5 – 82,5 for bin named 80, for example, so it is not perfect but does the job. From the graph we can see a concentration in the range of 77,5 to 87,5.

Part 4: Address and refine results

I consider the visualizations that I developed to be useful for examining different aspects of the coffee scores. One can combine looking at both dimensions, country and variety, with choosing different score types to concentrate on and then examining the rankings.

My overall impression from the data is that most of the coffees are of at least of rather good quality. This is demonstrated by the distribution of the total points. Also, when examining individual score types, one often encounters them ranging between approximately 7 and 8,5. The coffee producers submit samples themselves to be reviewed by the Coffee Quality Institute, so it is likely that the more confident the producer is of the quality, the more likely they are interested in submitting a sample.

Still one can observe certain consistency when looking at individual results. For total points the top three ones are United States, Papua New Guinea and Ethiopia. When selecting different categories, the African countries of Ethiopia, Uganda, Kenya and Rwanda are always in the top. This was something I expected, but I did not expect United States to be rated so highly. Brazil actually does not top any categories but is usually found in the middle of the pack.

When it comes to varieties, Ethiopian Yirgacheffe and Sumatran Lintong lead top points. Yellow Bourbon is around the mid-range.

Scoring of aftertaste reveals similar results. US, Ethiopia and Papua New Guinea are the top three. Same goes for varieties. Again, Ethiopian Yirgacheffe and Sumatran Lintong are in top 3 with SL28 from Kenya taking second place.

One thing to take into account is that there are big differences in how many samples are reviewed. Papua New Guinea has only one. There are few also for Ethiopian Yirgacheffe (2) and Sumatran Lintong (1). African varieties SL14, SL28 and SL34 have more samples, in the tens, and those varieties are found near top in different categories.

To sum up, the original questions:

Question 1: Which country in the world produces the best Arabica coffee

One can observe that quality coffee is produced all over the world. You cannot say for certainty that which country is the best one, but if one had to choose a region, then based on the data East African region of Ethiopia, Kenya, Uganda and Rwanda produces the best coffee alongside United States. The representatives of the US is questionable though with 8 samples.

Question 2: Which Arabica coffee variety is the best?

Based on data, one cannot confidently choose one variety. I assess the data so that again East Africa is a reliable choice to likely get top quality with emphasis on Kenyan SL14, SL28 and SL34 since they are better represented in the data.

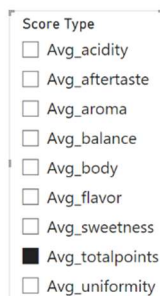
Question 3: When looking for coffees with a good aftertaste, which countries or varieties should one look for?

For aftertaste, the answer is similar than when it comes to overall total points. East Africa and US standout, although one cannot total exclude the high scores of Sumatran varieties, albeit with lower sample count.

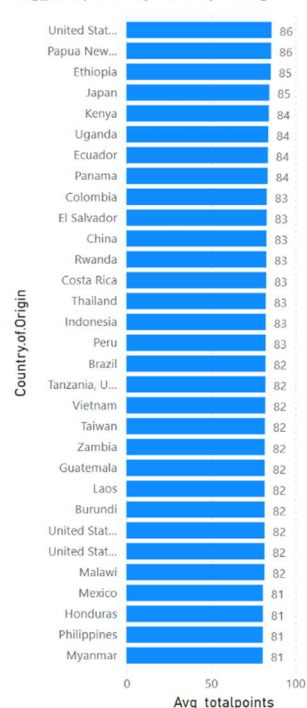
Part 5: Communicate Your Findings and Part 6: Track Outcomes

To support my earlier assessment, I chose these two visualizations. For total points:

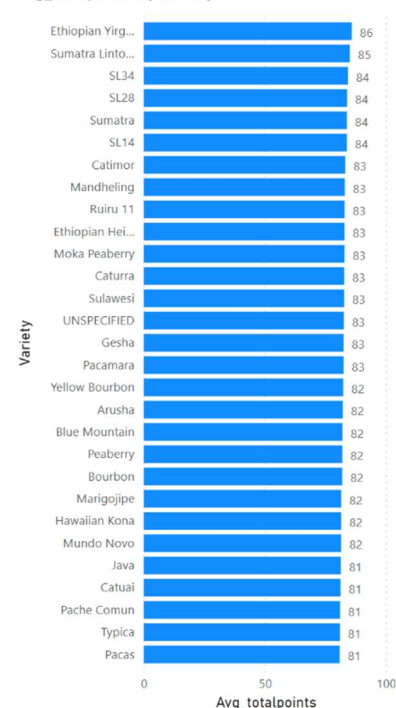
Avg_totalpoints by Country.of.Origin



Avg_totalpoints by Country.of.Origin



Avg_totalpoints by Variety



And aftertaste:

